In [1]:
```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

import warnings
warnings.filterwarnings("ignore")
```

In [3]:
```
data = pd.read_csv("global_cancer_patients_2015_2024 (1).csv")
```

In [4]:
```
data
```

Out[4]: Columns 1–8 of 16

	Patient_ID	Age	Gender	Country_Region	Year	Genetic_Risk	Air_Pollution
0	PT0000000	71	Male	UK	2021	6.4	2.8
1	PT0000001	34	Male	China	2021	1.3	4.5
2	PT0000002	80	Male	Pakistan	2023	7.4	7.9
3	PT0000003	40	Male	UK	2015	1.7	2.9
4	PT0000004	43	Female	Brazil	2017	5.1	2.8
...	...	...	...	...	...	...	...
49995	PT0049995	80	Male	Pakistan	2023	2.3	7.5
49996	PT0049996	40	Female	USA	2018	6.4	3.5
49997	PT0049997	74	Male	Pakistan	2015	6.2	1.6
49998	PT0049998	21	Male	Brazil	2018	4.0	6.5
49999	PT0049999	22	Male	Pakistan	2023	5.1	9.8

Columns 9–15 of 16 · first column repeated

	Alcohol_Use	Smoking	Obesity_Level	Cancer_Type	Cancer_Stage	Treatment_Cost_USD	Survival_Years
0	9.5	0.9	8.7	Lung	Stage III	62913.44	5.9
1	3.7	3.9	6.3	Leukemia	Stage 0	12573.41	4.7
2	2.4	4.7	0.1	Breast	Stage II	6984.33	7.1
3	4.8	3.5	2.7	Colon	Stage I	67446.25	1.6
4	2.3	6.7	0.5	Skin	Stage III	77977.12	2.9
...	...	...	...	...	...	...	...
49995	2.8	3.8	2.9	Leukemia	Stage 0	71463.70	1.0
49996	2.9	9.0	9.8	Colon	Stage I	49619.66	0.4
49997	8.7	4.7	4.0	Breast	Stage 0	5338.25	2.7
49998	7.6	8.6	8.1	Liver	Stage IV	45100.47	7.5
49999	3.2	0.0	0.7	Liver	Stage 0	64023.82	7.4

Columns 16–16 of 16 · first column repeated

	Target_Severity_Score
0	4.92
1	4.65
2	5.84
3	3.12
4	3.62
...	...
49995	3.63
49996	6.03

	Target_Severity_Score
49997	6.02
49998	6.54
49999	3.76

50000 rows × 15 columns

In [5]:
```
data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 15 columns):
 #   Column                 Non-Null Count  Dtype  
---  ------                 --------------  -----  
 0   Patient_ID             50000 non-null  str    
 1   Age                    50000 non-null  int64  
 2   Gender                 50000 non-null  str    
 3   Country_Region         50000 non-null  str    
 4   Year                   50000 non-null  int64  
 5   Genetic_Risk           50000 non-null  float64
 6   Air_Pollution          50000 non-null  float64
 7   Alcohol_Use            50000 non-null  float64
 8   Smoking                50000 non-null  float64
 9   Obesity_Level          50000 non-null  float64
 10  Cancer_Type            50000 non-null  str    
 11  Cancer_Stage           50000 non-null  str    
 12  Treatment_Cost_USD     50000 non-null  float64
 13  Survival_Years         50000 non-null  float64
 14  Target_Severity_Score  50000 non-null  float64
dtypes: float64(8), int64(2), str(5)
memory usage: 7.3 MB
```

In [6]:
```
data.duplicated().sum()
```

Out[6]:
```
np.int64(0)
```

Descriptive Analysis

In [8]:
```
data['Age']
```

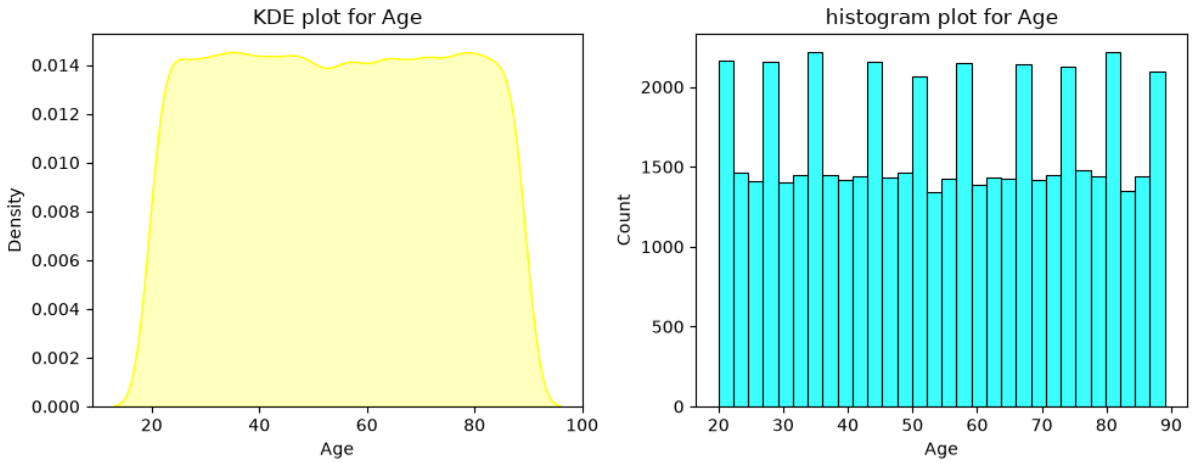
Out[8]:
```
0        71
1        34
2        80
3        40
4        43
         ..
49995    80
49996    40
49997    74
49998    21
49999    22
Name: Age, Length: 50000, dtype: int64
```

In [14]:
```
#plotting for age column
plt.figure(figsize=(10,4))

plt.subplot(1,2,1)
sns.kdeplot(data['Age'],fill=True,color='yellow')
plt.title("KDE plot for Age")

plt.subplot(1,2,2)
sns.histplot(data['Age'],bins=30,kde=False,color='cyan')
plt.title("histogram plot for Age")

plt.tight_layout()
plt.show()
```



In [15]:
```
data['Age'].describe()
```

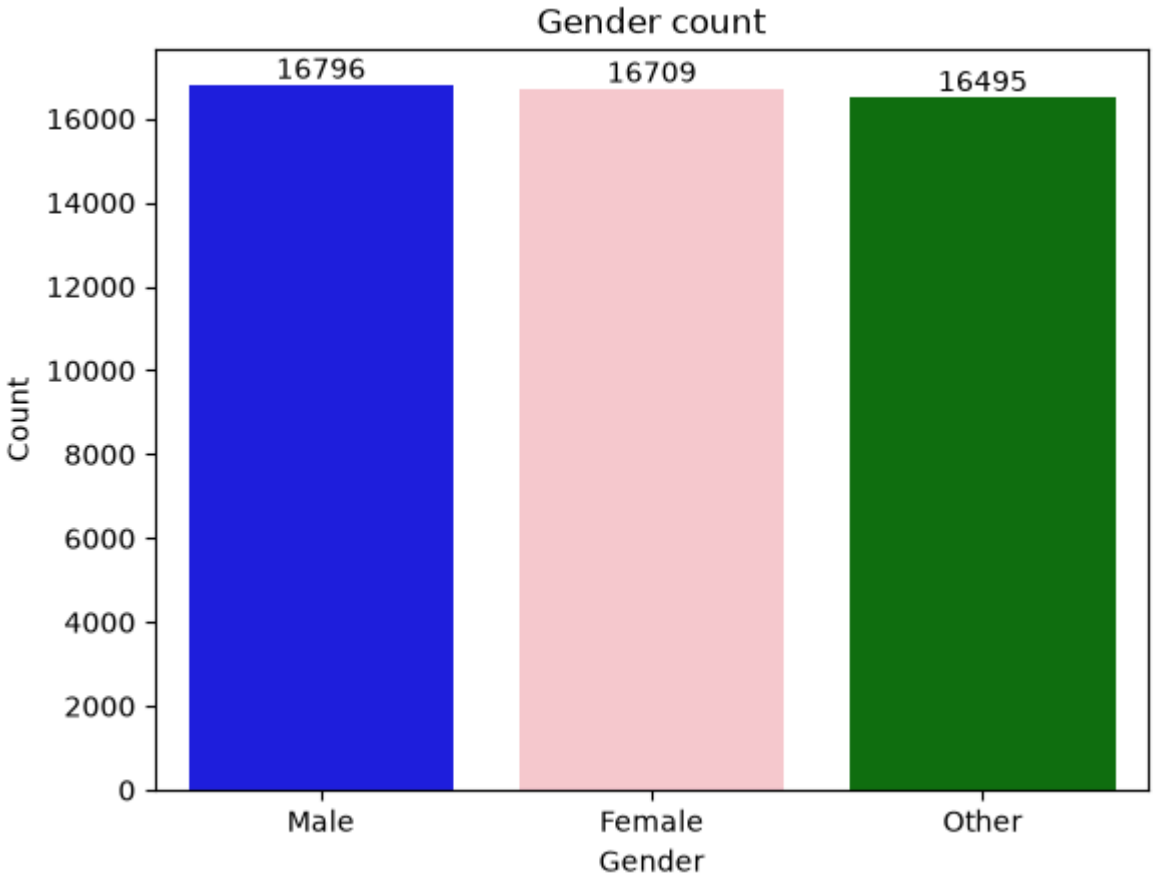
Out[15]:
```
count    50000.000000
mean        54.421540
std         20.224451
min         20.000000
25%         37.000000
50%         54.000000
75%         72.000000
max         89.000000
Name: Age, dtype: float64
```

In [16]:
```
# Gender Column
data['Gender'].value_counts()
```

Out[16]:
```
Gender
Male      16796
Female    16709
Other     16495
Name: count, dtype: int64
```

In [17]:
```
sns.barplot(x= data["Gender"].value_counts().index,
            y= data["Gender"].value_counts().values,
            palette=["blue", "pink", "green"]
           )
for i , v in enumerate(data["Gender"].value_counts()):
    plt.text(i,v, str(v), ha="center", va="bottom")

plt.title("Gender count")
plt.xlabel("Gender")
plt.ylabel("Count")
plt.show()
```



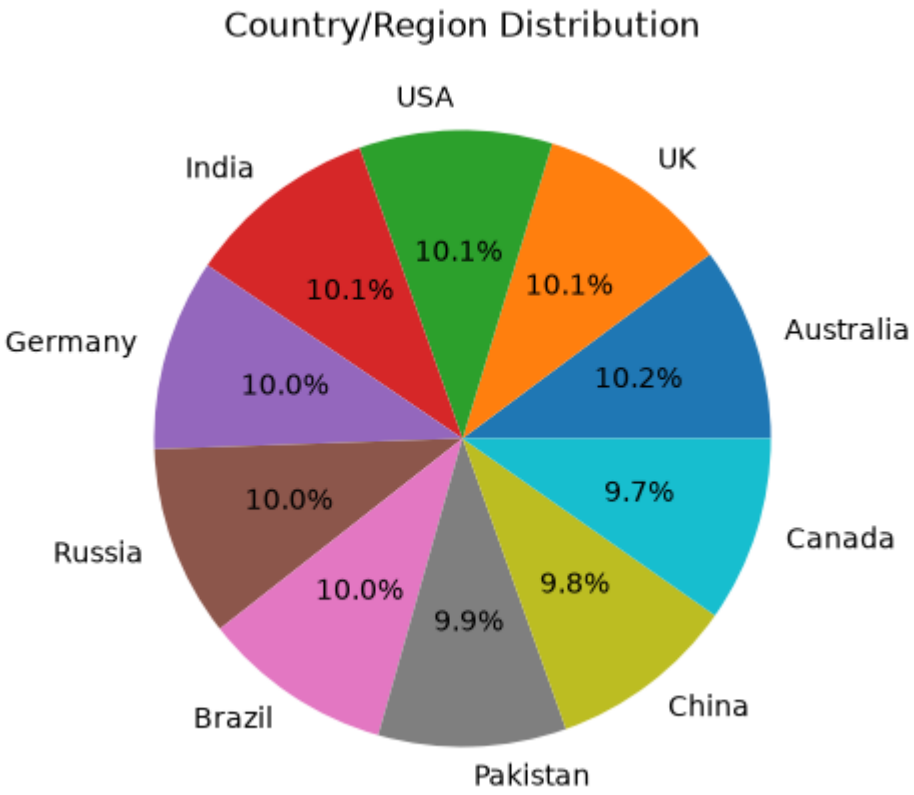
The dataset contain three gender categories(Male, Female, Other) with the most common being Male (16796 records).
Gender distribution is sufficient for evaluating gender-specific survivial trends and severity outcomes.

In [22]:
```
# countries column

country_counts = data['Country_Region'].value_counts()

plt.figure(figsize=(5,5))
plt.pie(x=country_counts.values,
        labels=country_counts.index,
        autopct= '%1.1f%%'
       )

plt.title("Country/Region Distribution")
plt.show()
```



In [23]:
```
country_counts
```

Out[23]:
```
Country_Region
Australia    5092
UK           5060
USA          5060
India        5040
Germany      5024
Russia       5017
Brazil       5004
Pakistan     4926
China        4913
Canada       4864
Name: count, dtype: int64
```

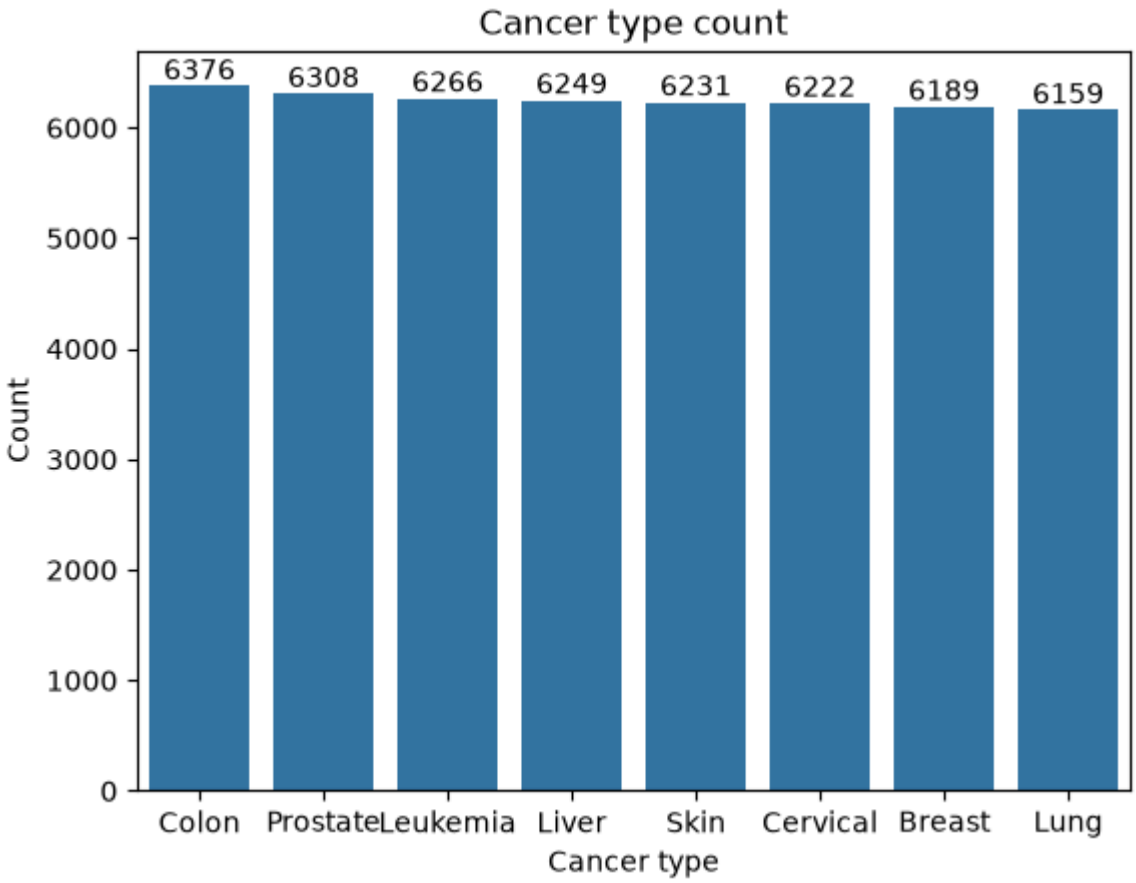
In [26]:
```
# cancer type
data["Cancer_Type"].value_counts()
```

Out[26]:
```
Cancer_Type
Colon       6376
Prostate    6308
Leukemia    6266
Liver       6249
Skin        6231
Cervical    6222
Breast      6189
Lung        6159
Name: count, dtype: int64
```

In [27]:
```
sns.barplot(x=data["Cancer_Type"].value_counts().index,
            y=data["Cancer_Type"].value_counts().values)

for i , v  in enumerate(data["Cancer_Type"].value_counts()):
    plt.text(i,v, str(v), ha="center", va="bottom")

plt.title("Cancer type count")
plt.xlabel("Cancer type")
plt.ylabel("Count")
plt.show()
```



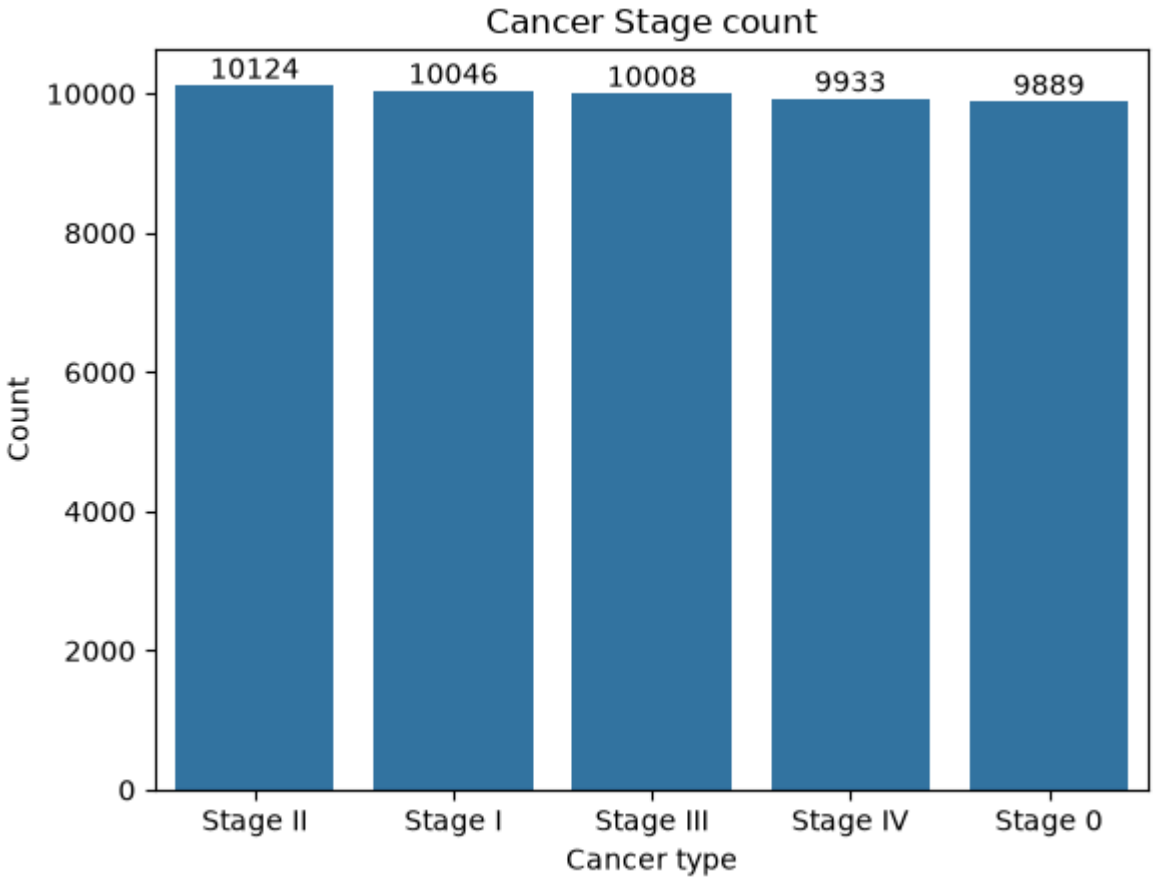
In [28]:
```
# Cancer Stage
data["Cancer_Stage"].value_counts()
```

Out[28]:
```
Cancer_Stage
Stage II     10124
Stage I      10046
Stage III    10008
Stage IV      9933
Stage 0       9889
Name: count, dtype: int64
```

In [29]:
```
sns.barplot(x=data["Cancer_Stage"].value_counts().index,
            y=data["Cancer_Stage"].value_counts().values)

for i , v  in enumerate(data["Cancer_Stage"].value_counts()):
    plt.text(i,v, str(v), ha="center", va="bottom")

plt.title("Cancer Stage count")
plt.xlabel("Cancer type")
plt.ylabel("Count")
plt.show()
```



In [30]:
```
data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 15 columns):
 #   Column                 Non-Null Count  Dtype  
---  ------                 --------------  -----  
 0   Patient_ID             50000 non-null  str    
 1   Age                    50000 non-null  int64  
 2   Gender                 50000 non-null  str    
 3   Country_Region         50000 non-null  str    
 4   Year                   50000 non-null  int64  
 5   Genetic_Risk           50000 non-null  float64
 6   Air_Pollution          50000 non-null  float64
 7   Alcohol_Use            50000 non-null  float64
 8   Smoking                50000 non-null  float64
 9   Obesity_Level          50000 non-null  float64
 10  Cancer_Type            50000 non-null  str    
 11  Cancer_Stage           50000 non-null  str    
 12  Treatment_Cost_USD     50000 non-null  float64
 13  Survival_Years         50000 non-null  float64
 14  Target_Severity_Score  50000 non-null  float64
dtypes: float64(8), int64(2), str(5)
memory usage: 7.3 MB
```

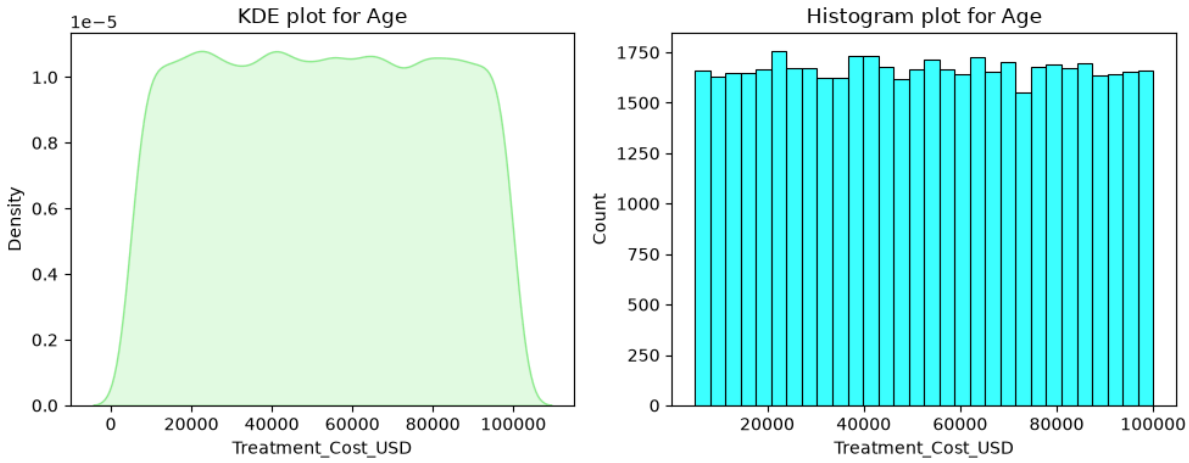
In [31]:
```
# plotting for treatment cost

# plotting for age column
plt.figure(figsize=(10,4))

plt.subplot(1,2,1)
sns.kdeplot(data["Treatment_Cost_USD"], fill=True, color="lightgreen")
plt.title("KDE plot for Age")

plt.subplot(1,2,2)
sns.histplot(data["Treatment_Cost_USD"], bins=30,  kde=False, color="cyan")
plt.title("Histogram plot for Age")

plt.tight_layout()
plt.show()
```



In [32]:
```
data["Treatment_Cost_USD"].describe()
```

Out[32]:
```
count    50000.000000
mean     52467.298239
std      27363.229379
min       5000.050000
25%      28686.225000
50%      52474.310000
75%      76232.720000
max      99999.840000
Name: Treatment_Cost_USD, dtype: float64
```

Treatment cost USD have no skewness , and there are almost same number of data points under each bin as observed by histogram

In [33]:
```
# ANALYLIZING THE RISK FACTORS
column_of_interest= ['Genetic_Risk','Air_Pollution', 'Alcohol_Use',
'Smoking', 'Obesity_Level']

summary= data[column_of_interest].agg(['mean', "std", "min", "max"])
summary
```

Out[33]:

	Genetic_Risk	Air_Pollution	Alcohol_Use	Smoking	Obesity_Level
mean	5.001698	5.010126	5.010880	4.989826	4.991176
std	2.885773	2.888399	2.888769	2.881579	2.894504
min	0.000000	0.000000	0.000000	0.000000	0.000000
max	10.000000	10.000000	10.000000	10.000000	10.000000

These variables have nearly identical means and standard deviations, indicating they were likely designed on the same standardized scale. They are essential in studying interaction effects (e.g., genetic risk × smoking) on survival.

Determine the relationship between risk factors and cancer

severity

In [35]:

```
from scipy.stats import linregress

risk_factors= ['Genetic_Risk','Air_Pollution', 'Alcohol_Use', 'Smoking',
'Obesity_Level']
titles= ['Genetic Risk','Air Pollution', 'Alcohol Use', 'Smoking', 'Obesity
Level']
colors= ["blue", "green", "orange", "red", "purple"]

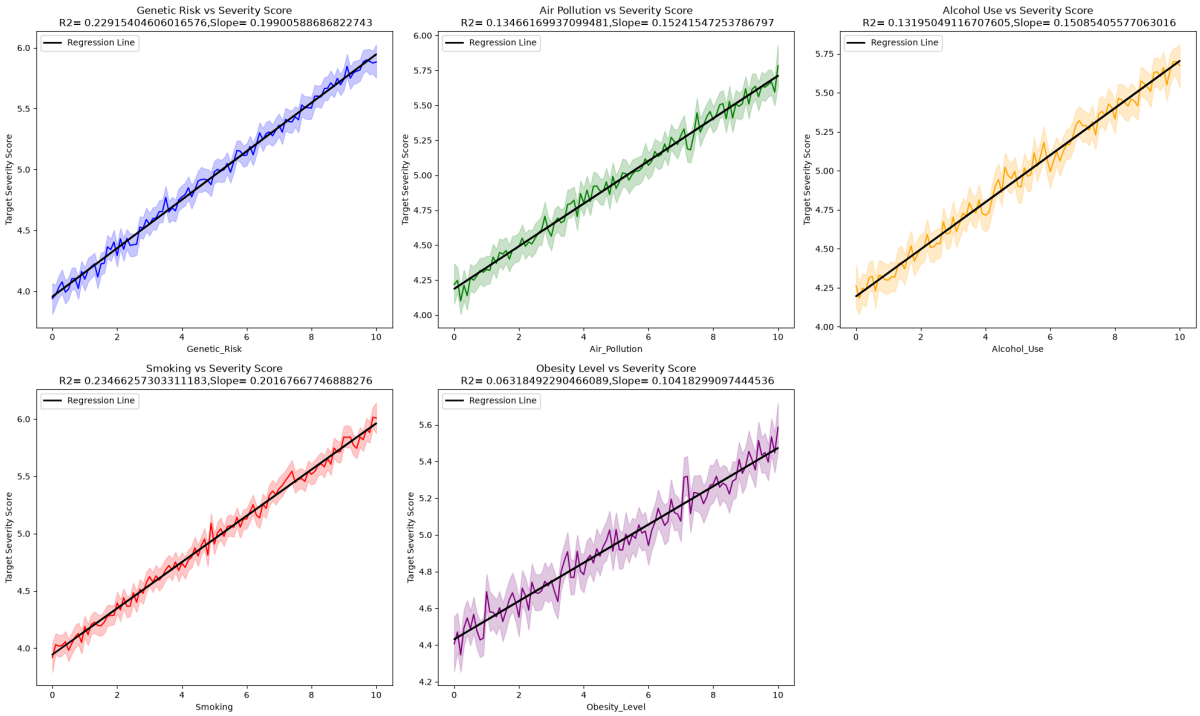
plt.figure(figsize=(20,12))
for i , (factor , title,color) in enumerate(zip(risk_factors, titles,
colors),1):
    plt.subplot(2,3,i)

    x=data[factor]
    y=data["Target_Severity_Score"]
    slope, intercept , r_value , p_value , std_err= linregress(x,y)
    r_squared= r_value**2

    sns.lineplot(x=factor, y="Target_Severity_Score", data=data,
color=color)
    plt.plot(x,x*slope+intercept, color="black", linewidth=2,
label="Regression Line")
    plt.title(f"{title} vs Severity Score\n R2= {r_squared},Slope= {slope}")
    plt.xlabel(factor)
    plt.ylabel("Target Severity Score")
    plt.legend()

#line= y = mx+b

plt.tight_layout()
plt.show()
```



To understand the contribution of various risk factors to cancer severity, line plots were generated for five primary variables: Genetic Risk, Air Pollution, Alcohol Use, Smoking, and Obesity Level, plotted against the Target Severity Score.
All graphs reveal a positive relationship, indicating that as the level of a particular risk factor increases, the corresponding severity of the condition also tends to rise. However, the degree of association—measured by the slope and tightness of the confidence interval—varies across factors.

Genetic Risk vs Target Severity Score
R^2 = 0.23:
A weak linear relationship. Only 23% of the variability in Target_Severity_Score can be explained by Genetic_Risk. This suggests that other factors likely play a larger role in influencing the severity score.
Slope = 0.20:
A positive slope indicates that as Genetic_Risk increases, the Target_Severity_Score also tends to increase. For each unit increase in Genetic_Risk, the target severity score increases by 0.20 units. However, because the R^2 is relatively low.

Air Pollution vs Target Severity Score
R^2 = 0.13:
A very weak relationship. Only 13% of the variance in Target_Severity_Score can be explained by Air_Pollution, meaning that this factor has a limited effect on the target variable.
Slope = 0.15:
A positive slope means that as air pollution increases, the severity score slightly increases. But, due to the low R^2, this relationship is weak and unreliable as a predictor for the target severity.

Alcohol Use vs Target Severity Score
R^2 = 0.13:
Similarly, the relationship between Alcohol_Use and Target_Severity_Score is also weak. Only 13% of the variation in the target score is explained by alcohol use.
Slope = 0.15:
The positive slope indicates that increased alcohol use correlates with a slight

increase in target severity. However, like air pollution, the weak R^2 suggests other factors have a much stronger influence on the target.

Smoking vs Target Severity Score
R^2 = 0.23:
A weak relationship, similar to Genetic_Risk. Smoking explains only 23% of the variance in the target score, leaving the majority of the variation to be explained by other factors.
Slope = 0.20:
The positive slope implies that as smoking increases, the target severity score increases as well. This relationship is similar to that of genetic risk, but with a weak linear association (low R^2).

Obesity Level vs Target Severity Score
R^2 = 0.06:
The weakest relationship among all factors. Only 6% of the variation in the target score is explained by obesity level, suggesting that obesity has a minimal effect on the target variable.
Slope = 0.10:
A positive slope, indicating a slight increase in the severity score as obesity level increases. However, due to the very low R^2, this is a weak and unreliable relationship.

Analyze the proportion of early-stage diagnoses by cancer type

In [37]:
```
data["Cancer_Type"].unique()
```

Out[37]:
```
<ArrowStringArray>
['Lung', 'Leukemia', 'Breast', 'Colon', 'Skin', 'Cervical', 'Prostate',
 'Liver']
Length: 8, dtype: str
```

In [38]:
```
stage_count= data[data["Cancer_Type"]=="Lung"]
["Cancer_Stage"].value_counts()
early_stage_sum= stage_count.get("Stage 0",0)+stage_count.get("Stage I",0)
total_sum= stage_count.sum()
proportion= (early_stage_sum/total_sum)*100
print(f"proportion  of lungs cancer  diagnosed  at stage 0 and Stag 1 :
{proportion}")
```

```
proportion  of lungs cancer  diagnosed  at stage 0 and Stag 1 :
38.43156356551388
```

In [39]:
```
stage_count= data[data["Cancer_Type"]=="Leukemia"]
["Cancer_Stage"].value_counts()
early_stage_sum= stage_count.get("Stage 0",0)+stage_count.get("Stage I",0)
total_sum= stage_count.sum()
proportion= (early_stage_sum/total_sum)*100
print(f"proportion  of Leukemia cancer  diagnosed  at stage 0 and Stag 1 :
{proportion}")
```

```
proportion  of Leukemia cancer  diagnosed  at stage 0 and Stag 1 :
39.53080114905841
```

In [40]:
```
stage_count= data[data["Cancer_Type"]=="Breast"]
["Cancer_Stage"].value_counts()
early_stage_sum= stage_count.get("Stage 0",0)+stage_count.get("Stage I",0)
total_sum= stage_count.sum()
proportion= (early_stage_sum/total_sum)*100
print(f"proportion  of Breast cancer  diagnosed  at stage 0 and Stag 1 :
{proportion}")
```

```
proportion  of Breast cancer  diagnosed  at stage 0 and Stag 1 :
39.47325900791727
```

In [41]:
```
stage_count= data[data["Cancer_Type"]=="Colon"]
["Cancer_Stage"].value_counts()
early_stage_sum= stage_count.get("Stage 0",0)+stage_count.get("Stage I",0)
total_sum= stage_count.sum()
proportion= (early_stage_sum/total_sum)*100
print(f"proportion  of Colon cancer  diagnosed  at stage 0 and Stag 1 :
{proportion}")
```

```
proportion  of Colon cancer  diagnosed  at stage 0 and Stag 1 :
40.41718946047679
```

In [43]:
```
stage_count= data[data["Cancer_Type"]=="Skin"]
["Cancer_Stage"].value_counts()
early_stage_sum= stage_count.get("Stage 0",0)+stage_count.get("Stage I",0)
total_sum= stage_count.sum()
proportion= (early_stage_sum/total_sum)*100
print(f"proportion  of Skin cancer  diagnosed  at stage 0 and Stag 1 :
{proportion}")
```

```
proportion  of Skin cancer  diagnosed  at stage 0 and Stag 1 :
40.41084898090194
```

In [44]:
```
stage_count= data[data["Cancer_Type"]=="Cervical"]
["Cancer_Stage"].value_counts()
early_stage_sum= stage_count.get("Stage 0",0)+stage_count.get("Stage I",0)
total_sum= stage_count.sum()
proportion= (early_stage_sum/total_sum)*100
print(f"proportion  of Cervical cancer  diagnosed  at stage 0 and Stag 1 :
{proportion}")
```

```
proportion  of Cervical cancer  diagnosed  at stage 0 and Stag 1 :
39.85856637737062
```

In [45]:
```
stage_count= data[data["Cancer_Type"]=="Prostate"]
["Cancer_Stage"].value_counts()
early_stage_sum= stage_count.get("Stage 0",0)+stage_count.get("Stage I",0)
total_sum= stage_count.sum()
proportion= (early_stage_sum/total_sum)*100
print(f"proportion  of Prostate cancer  diagnosed  at stage 0 and Stag 1 :
{proportion}")
```

```
proportion  of Prostate cancer  diagnosed  at stage 0 and Stag 1 :
40.187064045656314
```

In [46]:
```
stage_count= data[data["Cancer_Type"]=="Liver"]
["Cancer_Stage"].value_counts()
early_stage_sum= stage_count.get("Stage 0",0)+stage_count.get("Stage I",0)
total_sum= stage_count.sum()
proportion= (early_stage_sum/total_sum)*100
print(f"proportion  of Liver cancer  diagnosed  at stage 0 and Stag 1 :
{proportion}")
```

```
proportion  of Liver cancer  diagnosed  at stage 0 and Stag 1 :
40.614498319731155
```

The analysis demonstrates that early-stage diagnosis for various cancer types is relatively widespread, with most cancers having an early diagnosis rate between 38.43% and 40.61%. Liver Cancer shows the highest proportion, while Lung Cancer shows the lowest. These findings suggest that while screening and diagnostic methods are effective, improvements can still be made, particularly in lung cancer detection.

Further research into screening strategies, early intervention, and the use of advanced diagnostic technologies could help increase the proportion of early-stage diagnoses, ultimately leading to better survival rates and outcomes for cancer patients. The relatively small variations across the cancer types indicate that, in general, healthcare systems may need to focus on enhancing early detection uniformly, with targeted efforts to address specific gaps in detection, particularly for cancers like lung cancer.

Identify key predictors of cancer severity and survival years.

In [49]:
```
features= ["Age", 'Genetic_Risk','Air_Pollution', 'Alcohol_Use', 'Smoking',
'Obesity_Level']

targets= ["Survival_Years","Target_Severity_Score"]

#calculate correlations
pearson_corr= data[features+targets].corr(method="pearson")
spearman_corr= data[features+targets].corr(method="spearman")

# slice out only thr relationship with target variables
pearson_results= pearson_corr[targets]
spearman_results=spearman_corr[targets]

#combine both
correlation_df= pd.concat([pearson_results,spearman_results], axis=1, keys=
["Pearson","Spearman"])
correlation_df
```

Out[49]:

	Pearson		Spearman	
	Survival_Years	Target_Severity_Score	Survival_Years	Target_Severity_Score
Age	-0.000147	-0.001481	-0.000149	-0.002076
Genetic_Risk	0.001708	0.478700	0.001684	0.472193
Air_Pollution	0.000893	0.366963	0.000841	0.357528
Alcohol_Use	-0.000527	0.363250	-0.000543	0.354694
Smoking	0.001371	0.484420	0.001372	0.477881
Obesity_Level	0.009771	0.251366	0.009780	0.243152
Survival_Years	1.000000	0.004161	1.000000	0.003431
Target_Severity_Score	0.004161	1.000000	0.003431	1.000000

In [50]:
```
# random forest for target severity score
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import r2_score

# caonveting categorical columns to numerical columns
categorical_cols= ["Gender","Country_Region","Cancer_Type","Cancer_Stage"]
for col in categorical_cols:
    le= LabelEncoder()
    data[col]=le.fit_transform(data[col])

# Preparing  features and input
X= data.drop(columns=
["Patient_ID","Survival_Years","Target_Severity_Score","Treatment_Cost_USD"]
)
y_severity= data["Target_Severity_Score"]

# train test split
X_train_s, X_test_s, y_train_s, y_test_s= train_test_split(X, y_severity,
test_size=0.2, random_state=40)

# train the model
model= RandomForestRegressor(n_estimators=200, max_depth= None ,
min_samples_split=2,min_samples_leaf=1, random_state=40)
model.fit(X_train_s, y_train_s)

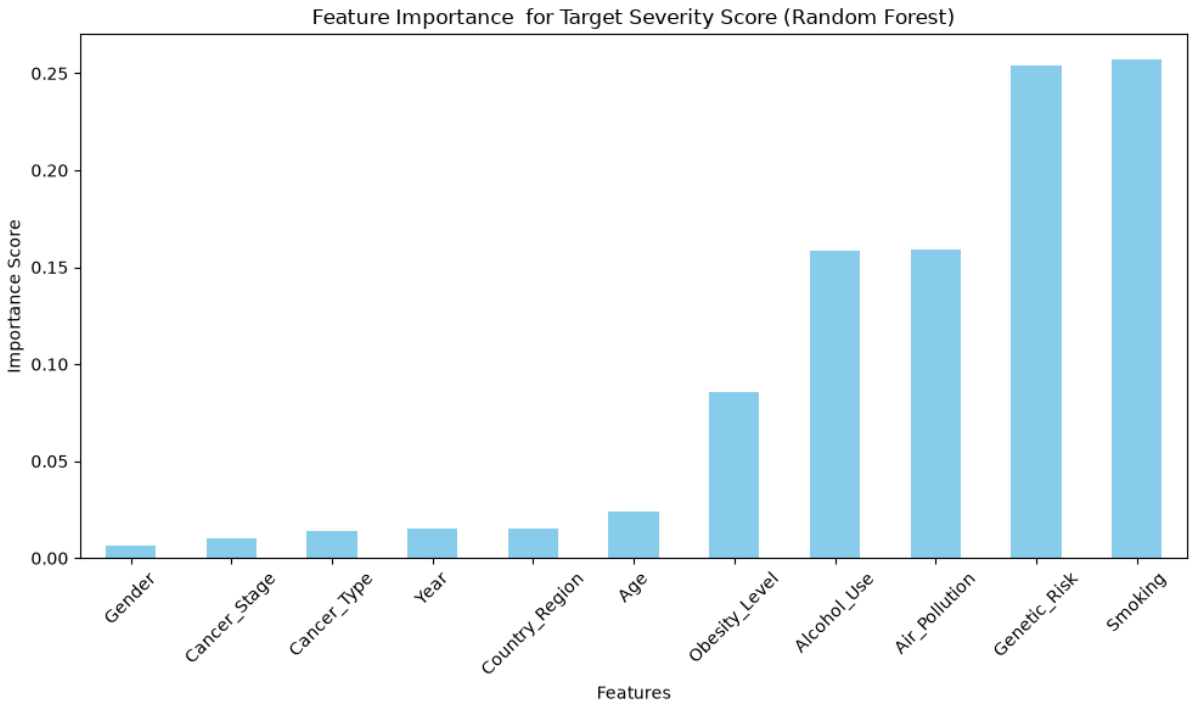
# evalaute the model
train_r2_severity= r2_score(y_train_s , model.predict(X_train_s))
test_r2_severity= r2_score(y_test_s , model.predict(X_test_s))
```

In [51]:
```
print(train_r2_severity)
print(test_r2_severity)
```

```
0.9689613345618926
0.7753533440404193
```

In [52]:
```
feature_importance_severity= pd.Series(model.feature_importances_,
index=X.columns,).sort_values(ascending=True)

#plotting of important features
plt.figure(figsize=(10,6))
feature_importance_severity.plot(kind="bar", color="skyblue")
plt.title("Feature Importance  for Target Severity Score (Random Forest)")
plt.xlabel("Features")
plt.ylabel("Importance Score")
plt.xticks(rotation= 45)
plt.tight_layout()
plt.show()
```



Feature Importance Interpretation
Smoking 0.2336 Most important predictor of severity score. The more a patient smokes, the higher their severity tends to be.
Genetic_Risk 0.2286 Strong genetic predisposition is nearly as important as smoking.
Treatment_Cost_USD 0.2133 Higher treatment costs are associated with more severe conditions.
Alcohol_Use 0.1291 Alcohol also plays a significant role.
Air_Pollution 0.1271 Environmental factor—patients in more polluted areas have worse severity scores.
Obesity_Level 0.0573 Has an effect, but much smaller.
Age to Gender < 0.01 Very low importance; these don’t explain much variation in severity score.

E.g., Smoking, Genetic Risk, AIr plollution are major influencers. This tells you where interventions might reduce severity.

In [55]:
```
# random forest for target severity score
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import r2_score

# caonveting categorical columns to numerical columns
categorical_cols= ["Gender","Country_Region","Cancer_Type","Cancer_Stage"]
for col in categorical_cols:
    le= LabelEncoder()
    data[col]=le.fit_transform(data[col])

# Preparing  features and input
X= data.drop(columns=
["Patient_ID","Survival_Years","Target_Severity_Score","Treatment_Cost_USD"]
)
y_severity= data["Survival_Years"]

# train test split
X_train_s, X_test_s, y_train_s, y_test_s= train_test_split(X, y_severity,
test_size=0.2, random_state=40)

param_grid = {
    'n_estimators': [100, 200],
    'max_depth': [5, 10, None],
    'min_samples_split': [2, 5],
    'min_samples_leaf': [1, 2]
}

# train the model
model= RandomForestRegressor(random_state=40)
GSC= GridSearchCV(model , param_grid, cv=3, scoring="r2", n_jobs=-1)
GSC.fit(X_train_s, y_train_s)

best_rf_severity= GSC.best_estimator_

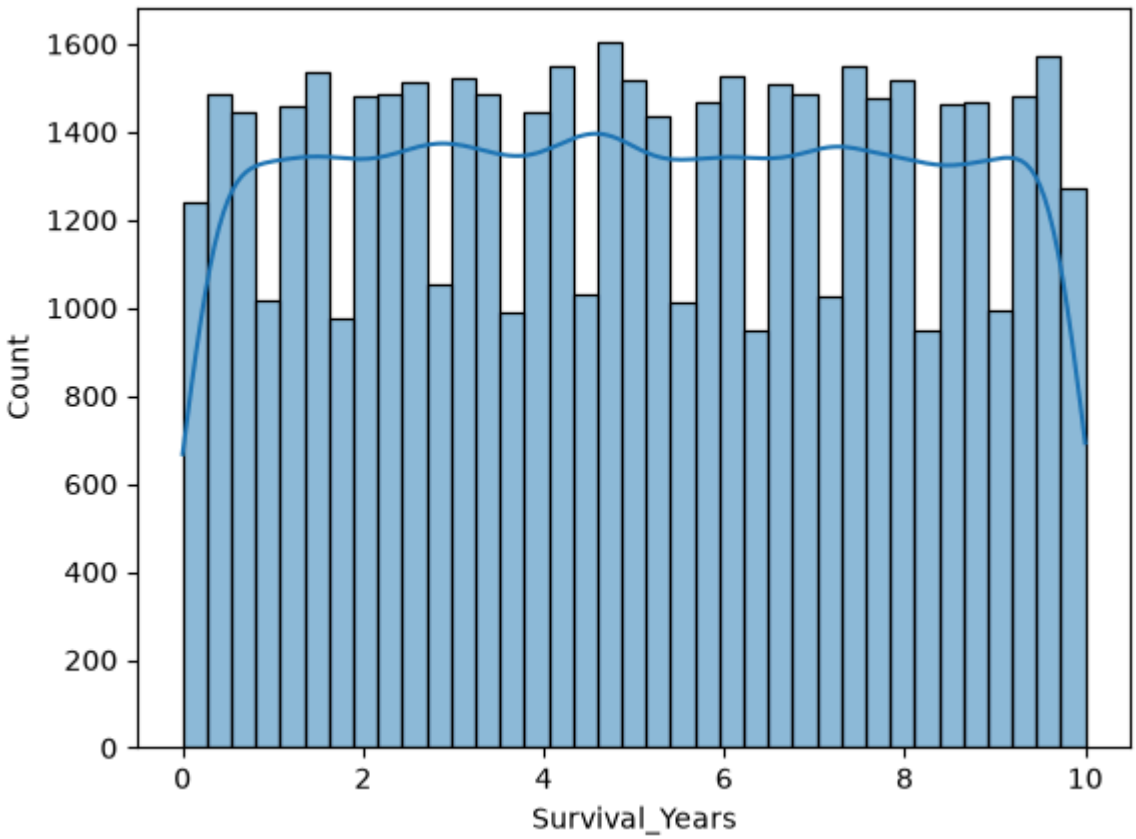
# evalaute the model
train_r2_severity= r2_score(y_train_s , best_rf_severity.predict(X_train_s))
test_r2_severity= r2_score(y_test_s , best_rf_severity.predict(X_test_s))
```

In [56]:
```
print(train_r2_severity)
print(test_r2_severity)
```

```
0.008998065861791527
-0.0003487352301700941
```

In [57]:
```
import seaborn as sns
sns.histplot(data["Survival_Years"], kde=True)
```

Out[57]:
```
<Axes: xlabel='Survival_Years', ylabel='Count'>
```



In [58]:
```
data.corr(numeric_only=True)["Survival_Years"].sort_values(ascending=True)
```

Out[58]:
```
Gender                  -0.007121
Cancer_Stage            -0.003014
Cancer_Type             -0.001717
Alcohol_Use             -0.000527
Treatment_Cost_USD      -0.000429
Age                     -0.000147
Year                     0.000018
Air_Pollution            0.000893
Smoking                  0.001371
Genetic_Risk             0.001708
Target_Severity_Score    0.004161
Country_Region           0.005121
Obesity_Level            0.009771
Survival_Years           1.000000
Name: Survival_Years, dtype: float64
```

"The information in your columns does not help the model figure out how long someone will survive."

Explore the economic burden of cancer treatment across different demographics and

countries

In [60]:
```
data = pd.read_csv("global_cancer_patients_2015_2024 (1).csv")
data
```

Out[60]: Columns 1–8 of 16

	Patient_ID	Age	Gender	Country_Region	Year	Genetic_Risk	Air_Pollution
0	PT0000000	71	Male	UK	2021	6.4	2.8
1	PT0000001	34	Male	China	2021	1.3	4.5
2	PT0000002	80	Male	Pakistan	2023	7.4	7.9
3	PT0000003	40	Male	UK	2015	1.7	2.9
4	PT0000004	43	Female	Brazil	2017	5.1	2.8
...	...	...	...	...	...	...	...
49995	PT0049995	80	Male	Pakistan	2023	2.3	7.5
49996	PT0049996	40	Female	USA	2018	6.4	3.5
49997	PT0049997	74	Male	Pakistan	2015	6.2	1.6
49998	PT0049998	21	Male	Brazil	2018	4.0	6.5
49999	PT0049999	22	Male	Pakistan	2023	5.1	9.8

Columns 9–15 of 16 · first column repeated

	Alcohol_Use	Smoking	Obesity_Level	Cancer_Type	Cancer_Stage	Treatment_Cost_USD	Survival_Years
0	9.5	0.9	8.7	Lung	Stage III	62913.44	5.9
1	3.7	3.9	6.3	Leukemia	Stage 0	12573.41	4.7
2	2.4	4.7	0.1	Breast	Stage II	6984.33	7.1
3	4.8	3.5	2.7	Colon	Stage I	67446.25	1.6
4	2.3	6.7	0.5	Skin	Stage III	77977.12	2.9
...	...	...	...	...	...	...	...
49995	2.8	3.8	2.9	Leukemia	Stage 0	71463.70	1.0
49996	2.9	9.0	9.8	Colon	Stage I	49619.66	0.4
49997	8.7	4.7	4.0	Breast	Stage 0	5338.25	2.7
49998	7.6	8.6	8.1	Liver	Stage IV	45100.47	7.5
49999	3.2	0.0	0.7	Liver	Stage 0	64023.82	7.4

Columns 16–16 of 16 · first column repeated

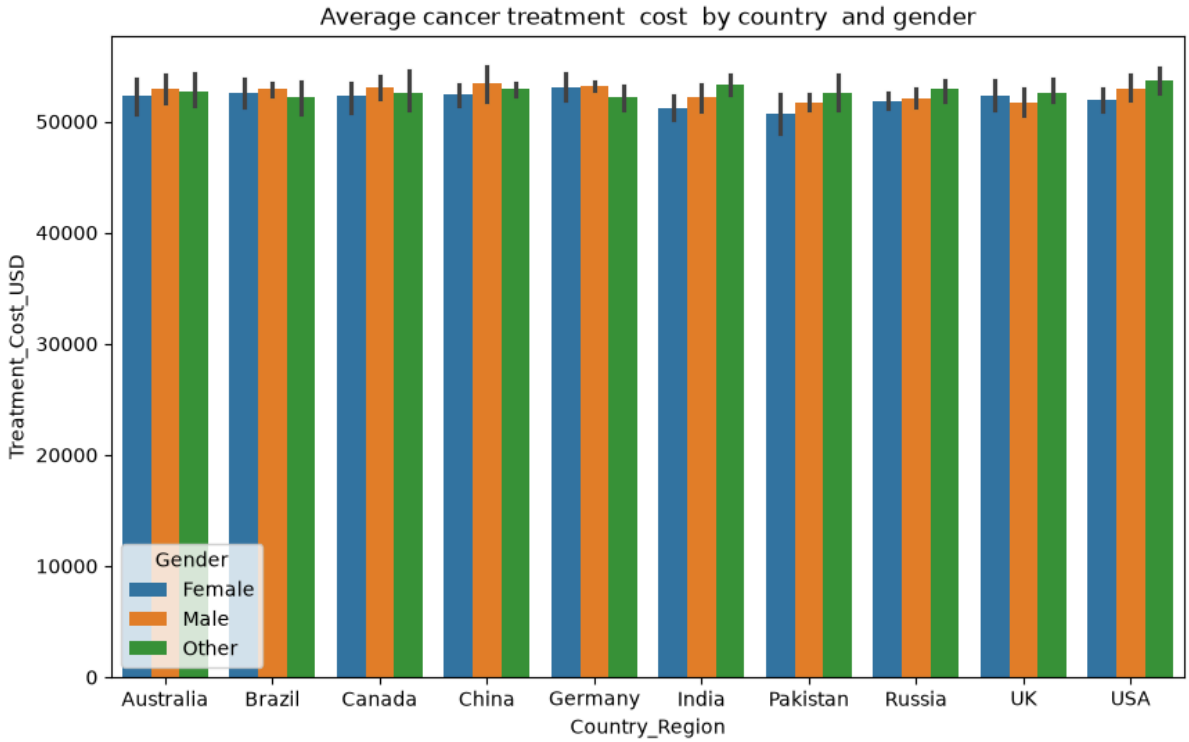
	Target_Severity_Score
0	4.92
1	4.65
2	5.84
3	3.12
4	3.62
...	...
49995	3.63
49996	6.03

	Target_Severity_Score
49997	6.02
49998	6.54
49999	3.76

50000 rows × 15 columns

In [61]:
```
data["Age_Group"]=pd.cut(data["Age"],bins=[0,30,45,60,75,100],labels=["0-30","31-45","46-60","61-75","76+"])
```

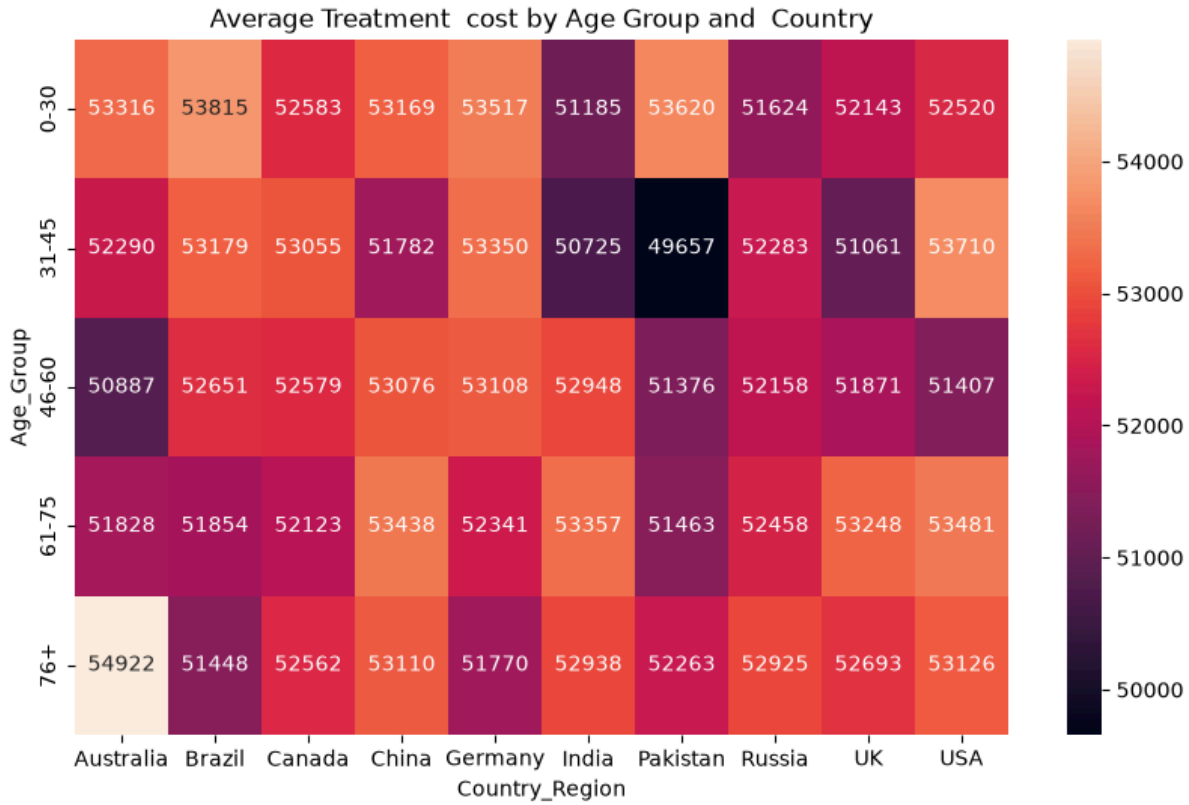
In [62]:
```
country_age_cost= data.groupby(["Country_Region", "Age_Group","Gender"])["Treatment_Cost_USD"].mean().reset_index()
plt.figure(figsize=(10,6))
sns.barplot(data=country_age_cost,x="Country_Region", y="Treatment_Cost_USD", hue="Gender")
plt.title("Average cancer treatment  cost  by country  and gender")
plt.show()
```



In [63]:
```
country_age_cost= data.groupby(["Country_Region", "Age_Group"])["Treatment_Cost_USD"].mean().reset_index()
```

In [64]:
```
heatmap_data= country_age_cost.pivot(index="Age_Group", columns="Country_Region", values="Treatment_Cost_USD")
```

In [65]:
```
plt.figure(figsize=(10,6))
sns.heatmap(heatmap_data, annot=True, fmt=".0f")
plt.title("Average Treatment  cost by Age Group and  Country")
plt.show()
```



Geographic Disparities in Economic Burden
Cancer treatment costs are significantly higher in developed nations such as the USA, Australia, and China, revealing the heavy financial load in advanced healthcare systems. Meanwhile, countries like India and Pakistan exhibit comparatively lower costs, likely due to lower healthcare pricing structures or limited access to advanced treatment. This highlights a clear global inequality in healthcare affordability that can intensify financial strain depending on a patient's country of residence.

Gender-Based Cost Patterns Are Uniform
Across all countries, gender-based differences in average treatment costs are minimal, suggesting no major gender bias in pricing or access to cancer care. This uniformity may reflect standardization in treatment protocols or equitable healthcare policies, but it also points to the fact that the financial impact of cancer is universal across genders.

Age-Related Escalation in Treatment Costs
Treatment costs tend to rise with age, particularly for those aged 61 and above. This trend is especially evident in countries like Australia and the USA, where older age groups face sharply higher costs. The increased financial burden in these groups could be due to more intensive care needs, multiple comorbidities, or prolonged treatments. This pattern underlines the vulnerability of elderly populations and the pressing need for targeted support for senior citizens.

Role of Healthcare Systems in Cost Variation

Countries with robust public healthcare systems—such as Canada, Germany, and the UK—show relatively stable treatment costs across age groups, reflecting the benefits of healthcare subsidies or coverage. This consistency reinforces the importance of government intervention and universal healthcare in mitigating financial disparities in cancer treatment.

Assess whether higher treatment cost is associated with longer survival

Null Hypothesis (H_0) :- There is no correlation between treatment cost and survival years.
Alternative Hypothesis (H_1):- There is a correlation (positive or negative) between treatment cost and survival years.

In [66]:
```
from scipy.stats  import pearsonr, spearmanr
x= data["Treatment_Cost_USD"]
y=data["Survival_Years"]

# performing pearson correlation test
pearson_corr, pearson_p = pearsonr(x,y)
print(f"Pearson Correlation Coefficient: {pearson_corr}")
print(f"Pearson P-Value : {pearson_p}")

# Spearman correlation test
spearman_corr, spearman_p = spearmanr(x,y)
print(f"spearman Correlation Coefficient: {spearman_corr}")
print(f"spearman P-Value : {spearman_p}")

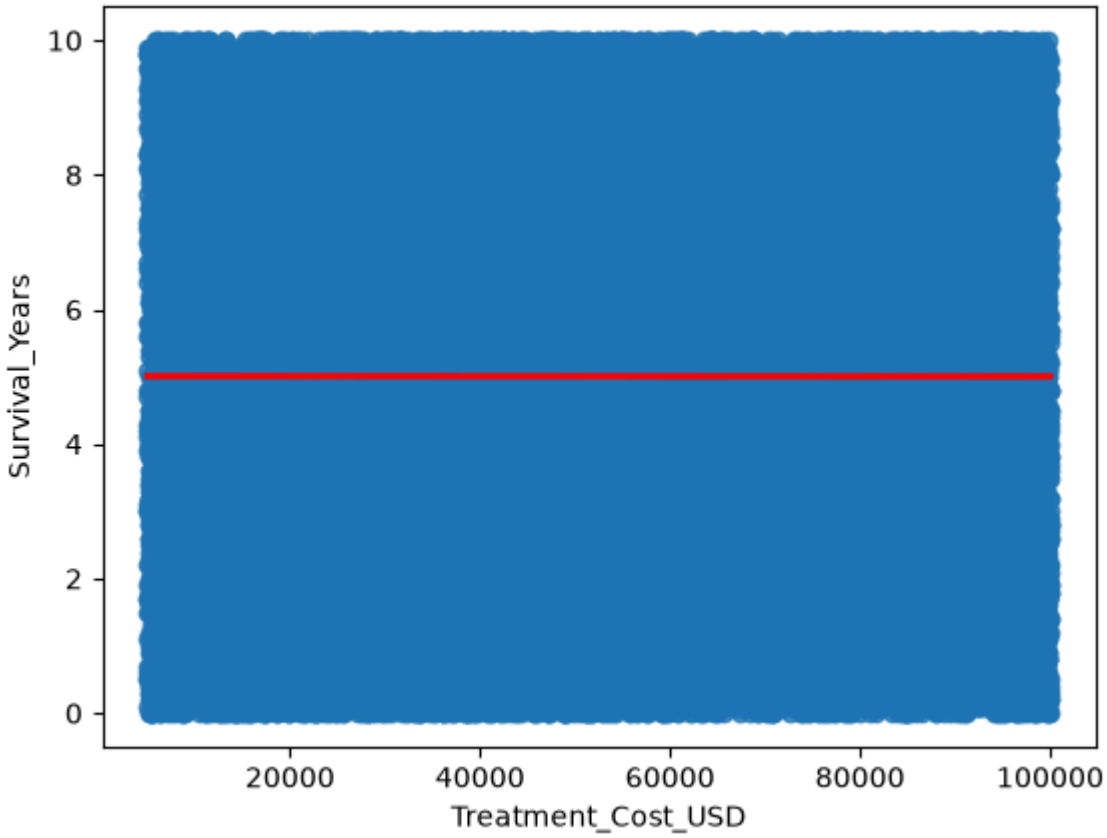
alpha=0.05

def interpret_corr(corr, p, method):
    if p<alpha:
        print(f"{method}, we  reject thr hull hypothesis")
    else:
        print(f"{method}, we failed to reject null hypothesis")

interpret_corr(pearson_corr,pearson_p,"Pearson")
interpret_corr(spearman_corr,spearman_p,"Spearman")
```

```
Pearson Correlation Coefficient: -0.00042940542830214906
Pearson P-Value : 0.9235082012963338
spearman Correlation Coefficient: -0.0004454737292549569
spearman P-Value : 0.9206551702014917
Pearson, we failed to reject null hypothesis
Spearman, we failed to reject null hypothesis
```

In [67]:
```
sns.regplot(x=x, y=y, line_kws={"color":"red"})
plt.show()
```



There is no relationship between Treatment cost and survival years.

Evaluate if higher cancer stages lead to greater treatment costs and reduced survival years

In [68]:
```
df= data.copy()
```

In [69]:
```
stage_order = ['Stage 0', 'Stage I', 'Stage II', 'Stage III', 'Stage IV']
```

In [70]:
```
grouped_stats= df.groupby("Cancer_Stage")
[["Treatment_Cost_USD","Survival_Years"]].mean().reset_index()
```

In [71]:
```
grouped_stats
```

Out[71]:

	Cancer_Stage	Treatment_Cost_USD	Survival_Years
0	Stage 0	52572.589493	5.015199
1	Stage I	52674.079638	5.012990
2	Stage II	52082.841258	4.995170
3	Stage III	52708.197506	5.036311
4	Stage IV	52302.471041	4.972596

Treatment Cost vs. Cancer Stage
Null Hypothesis (H_0): The average treatment cost is the same across all cancer stages.
Alternative Hypothesis (H_1): At least one stage has a different average cost.

Survival Years vs. Cancer Stage
Null Hypothesis (H_0): The average survival years are the same across all cancer stages.
Alternative Hypothesis (H_1): At least one stage has a different survival duration.

In [73]:
```
grouped_costs=[]
grouped_survival=[]


for  stage in stage_order:
    stage_data= df[df["Cancer_Stage"]==stage]
    cost= stage_data["Treatment_Cost_USD"]
    survival= stage_data["Survival_Years"]
    grouped_costs.append(cost)
    grouped_survival.append(survival)
```

In [74]:
```
len(grouped_costs)
```

Out[74]:
```
5
```

In [76]:
```
# check for normaility
from scipy.stats import  shapiro, f_oneway
normal_cost=0
normal_survival=0

for i in range (len(stage_order)):
    cost_p= shapiro(grouped_costs[i]).pvalue
    surv_p= shapiro(grouped_survival[i]).pvalue
    print(f" cost {cost_p} for group {i}")
    print(f"Survival {surv_p} for group {i}")
    if cost_p<0.05:
        normal_cost+=1
    if surv_p<0.05:
        normal_survival+=1
```

```
 cost 1.547183492964763e-47 for group 0
Survival 1.629105417956777e-47 for group 0
 cost 7.306530551009064e-48 for group 1
Survival 7.575557907296775e-48 for group 1
 cost 8.943587880475871e-48 for group 2
Survival 4.041663138012407e-48 for group 2
 cost 5.2672747384297505e-48 for group 3
Survival 1.3648532948152406e-47 for group 3
 cost 1.583865447110653e-47 for group 4
Survival 2.299261925745223e-47 for group 4
```

In [77]:
```
print(normal_cost)
print(normal_survival)
```

```
5
5
```

In [78]:
```
from scipy.stats import kruskal
```

In [79]:
```
kusrkal_cost =kruskal(*grouped_costs)
kurkal_survival= kruskal(*grouped_survival)

p_cost= kusrkal_cost.pvalue
p_survival= kurkal_survival.pvalue
```

In [80]:
```
p_cost
```

Out[80]:
```
np.float64(0.4254451689062656)
```

In [81]:
```
p_survival
```

Out[81]:
```
np.float64(0.6033078729984223)
```

```
=== Kruskal-Wallis Test: Treatment Cost across Cancer Stages ===

P-value: 0.4254
Conclusion: No significant difference in treatment costs among cancer stages.

=== Kruskal-Wallis Test: Survival Years across Cancer Stages ===

P-value: 0.6033
Conclusion: No significant difference in survival years among cancer stages.
```

Examine whether higher genetic risk amplifies the negative effects of smoking on cancer severity

```
Null Hypothesis (H₀):
The interaction effect between genetic risk and smoking on cancer Severity is not
significant.
(Genetic risk does not amplify or alter the effect of smoking.)

Alternative Hypothesis (H₁):
The interaction effect between genetic risk and smoking on  cancer severity is
significant.
(Genetic risk does amplify or alter the effect of smoking.)
```

In [87]:
```
import statsmodels.formula.api as smf

model = smf.ols("Target_Severity_Score ~ Genetic_Risk*Smoking",
data=data).fit()

model.summary2().tables[1].loc["Genetic_Risk:Smoking"]
```

Out[87]:
```
Coef.      -0.000228
Std.Err.    0.000471
t          -0.484187
P>|t|       0.628255
[0.025     -0.001150
0.975]      0.000694
Name: Genetic_Risk:Smoking, dtype: float64
```

In [85]:
```
p_value=0.628255
```

Interpretation:
The interaction coefficient is negative but very small: -0.000228

The p-value = 0.628, which is greater than 0.05, so we fail to reject the null hypothesis

🧪 Conclusion (Statistical):
✅ Test Used: Multiple Linear Regression with interaction term

📌 Null Hypothesis (H_0): No interaction effect between genetic risk and smoking

📌 Alternative Hypothesis (H_1): There is an interaction effect

The interaction effect between Genetic Risk and Smoking on Target Severity Score is not statistically significant (p = 0.628 > 0.05).

This means that based on your data, there is no evidence that Genetic Risk amplifies or reduces the effect of Smoking on the Target Severity Score.

In other words, smoking and genetic risk may each have independent effects (or none), but they do not interact in a way that significantly changes the outcome.